

COSC 39: Theory of Computation

Credit Statement: Talked to Carson Bates'27, Jack Bolitho'27, and Sair Shaikh'26.

Problem 1. Prove that all following languages are undecidable.

1. Prove that there is no Turing machine that decides the language

$$\text{EQUAL} := \{(\langle M_1 \rangle, \langle M_2 \rangle) \mid L(M_1) = L(M_2) \text{ for Turing machine } M_1 \text{ and } M_2\}$$

2. Prove that there is no Turing machine that decides the language

$$\text{REG} := \{\langle M \rangle \mid L(M) \text{ is regular}\}$$

Solution.

1. Assume for contradiction that some Turing machine M_{EQ} decides EQUAL. We will reduce the uncomputable problem

$$\text{NEVERACCEPT} := \{\langle M \rangle \mid \text{Turing machine } M \text{ never accepts any input}\}$$

to EQUAL. Let $M_\emptyset$ be a fixed Turing machine that immediately rejects every input (so $L(M_\emptyset) = \emptyset$ by construction). Construct a wrapper function WRAP around M_{EQ} : Given an input $\langle M \rangle$, WRAP proceeds as follows:

- (a) Run $M_{EQ}(\langle M \rangle, \langle M_\emptyset \rangle)$.
- (b) Return the result directly.

Then,

- If M accepts no input, we have $L(M) = \emptyset = L(M_\emptyset)$. As a result, M_{EQ} accepts $(\langle M \rangle, \langle M_\emptyset \rangle)$, and thus $\langle M \rangle \in L(\text{NEVERACCEPT})$.
- If M accepts some input, we have $L(M) \neq \emptyset = L(M_\emptyset)$. As a result, M_{EQ} rejects $(\langle M \rangle, \langle M_\emptyset \rangle)$, and thus $\langle M \rangle \notin L(\text{NEVERACCEPT})$.

Thus, WRAP decides NEVERACCEPT, which is a contradiction and hence no such M_{EQ} can exist that decides EQUAL.

2. Assume for contradiction that some Turing machine M_{REG} decides REG. We will reduce the uncomputable problem

$$\text{ACCEPT} := \{\langle M, w \rangle \mid \text{Turing machine } M \text{ accepts } w\}$$

to REG. We construct a Turing machine M' as follows, given input x :

- (a) Accept if $x \in 0^n 1^n$, $n \geq 1$.
- (b) If $x \notin 0^n 1^n$, simulate M with input w and accept if M accepts w .
- (c) Reject otherwise.

We then use the encoding of M' , $\langle M' \rangle$ as input for M_{REG} . Then we have the following cases,

- If M accepts w , $L(M') = \Sigma^*$. Thus $L(M')$ is regular and M_{REG} accepts.
- If M does not accept w , $L(M') = \{0^n 1^n \mid n \geq 1\}$. Thus $L(M')$ is not regular and M_{REG} rejects.

Thus, M_{REG} decides ACCEPT, which is a contradiction and hence no such M_{REG} can exist that decides REG.